
AI-Based Construction Site Surveillance System

An End-to-End Multi-Model Computer Vision System for Real-Time Construction Site Safety Monitoring, Predictive Risk Analysis, and Automated Emergency Alerting

CATEGORY · ARTIFICIAL INTELLIGENCE / COMPUTER VISION — APPLIED DEEP LEARNING FOR INDUSTRIAL SAFETY

Prepared as a project report documenting system design, models, engineering methodology, frontend application, and real-world impact.

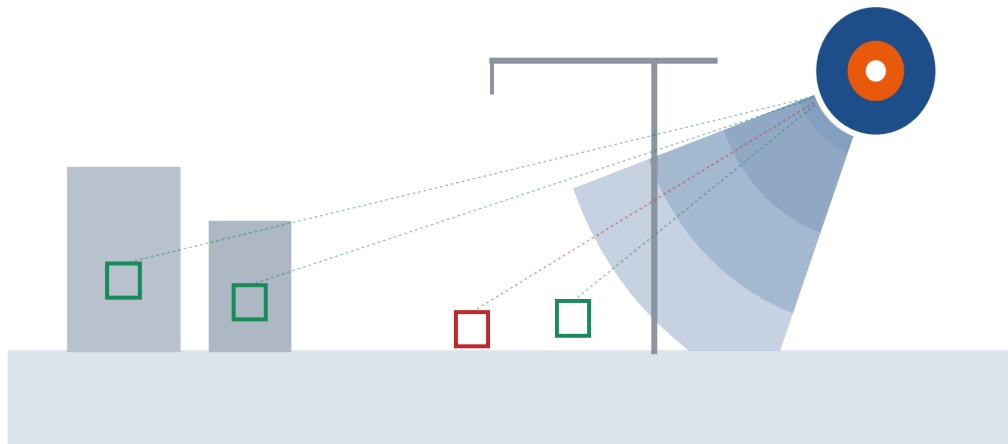


Figure 0 — Conceptual overview: AI-driven detection across a monitored construction site.

This document covers: project overview & motivation · system architecture · the application interface · sample detection output · datasets & pretrained models · the AI processing pipeline · the alarm & email notification system · the frontend dashboard · a full Product Requirements Document · real-world impact · technology stack · limitations, future scope, and conclusion.

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1. Executive Summary

AI-Based Construction Site Surveillance System is an end-to-end, multi-model computer vision system that continuously monitors construction site footage to detect safety violations and emergencies in real time, alert site supervisors instantly, and provide predictive risk insights before incidents escalate. The system was engineered in two coordinated parts: a GPU-accelerated AI detection pipeline (developed and benchmarked on Kaggle) and a purpose-built web monitoring dashboard (developed in Visual Studio Code) that makes the pipeline's output usable by non-technical site staff.

The system integrates six detection capabilities, backed by genuine pretrained deep learning models rather than simulated or placeholder logic, and pairs this pipeline with an automated alarm and email-notification system.

Core Capabilities

- **PPE Compliance Detection** — identifies whether workers are wearing helmets and safety vests.
- **Fire & Smoke Detection** — flags active fire or smoke events on camera in real time.
- **Fight / Altercation Detection** — detects physical altercations between individuals on site.
- **Fallen / Inactive Worker Detection** — identifies workers who have collapsed or stopped moving for an unsafe duration.
- **Live Worker Headcount** — continuously counts workers visible in frame and tracks unique individuals across a video.
- **Structural Anomaly Detection** — a rule-based tripwire flagging sudden, large structural changes that may indicate a collapse.
- **Automated Alarm & Email Alerts** — triggers a visual/audible alarm and emails an annotated screenshot to the site owner the moment a critical event occurs.
- **Predictive Risk Scoring** — aggregates recent violations and hazards into a live site risk score.

2. Problem Statement and Motivation

Construction remains one of the most hazardous industries in the world. According to U.S. Bureau of Labor Statistics data, construction workers account for roughly one in five workplace fatalities nationally, with a fatality rate several times higher than the all-industry average. OSHA's well-documented "Fatal Four" hazards — falls, struck-by incidents, electrocutions, and caught-in/between accidents — together account for a large majority of construction deaths each year, and falls alone remain the single largest cause, responsible for several hundred deaths annually in the U.S.

A significant share of these incidents share a common root cause: a hazard or violation existed and was visible on camera, but no one was watching at the moment it mattered. Manual monitoring does not scale — a human supervisor cannot simultaneously watch every camera feed, every worker, and every corner of an active site, and fatigue and attention drift make missed incidents inevitable over an 8-to-12-hour shift.

AI-Based Construction Site Surveillance System was built to close this gap: to provide continuous, automated, unbiased monitoring that never looks away, flags danger the moment it appears, and gets the right person's

attention immediately rather than after the fact.

Sources: U.S. Bureau of Labor Statistics fatal occupational injury data; Occupational Safety and Health Administration (OSHA) "Fatal Four" construction safety guidance.

3. System Architecture

Input footage (an uploaded video or image today, with live-camera ingestion planned as a future extension) is first normalized through FFmpeg to guarantee compatibility with a wide range of video codecs and containers — a necessary step, since raw footage from different cameras and phones frequently uses codecs that standard vision libraries cannot reliably decode. Frames are then passed in parallel through five deep learning detectors and one rule-based structural monitor. Detections are merged by an aggregation and overlay engine, which draws bounding boxes, renders the live worker-count HUD, raises alert banners, and produces an annotated output video alongside a structured, timestamped event log used for analytics, reporting, and the alarm system.

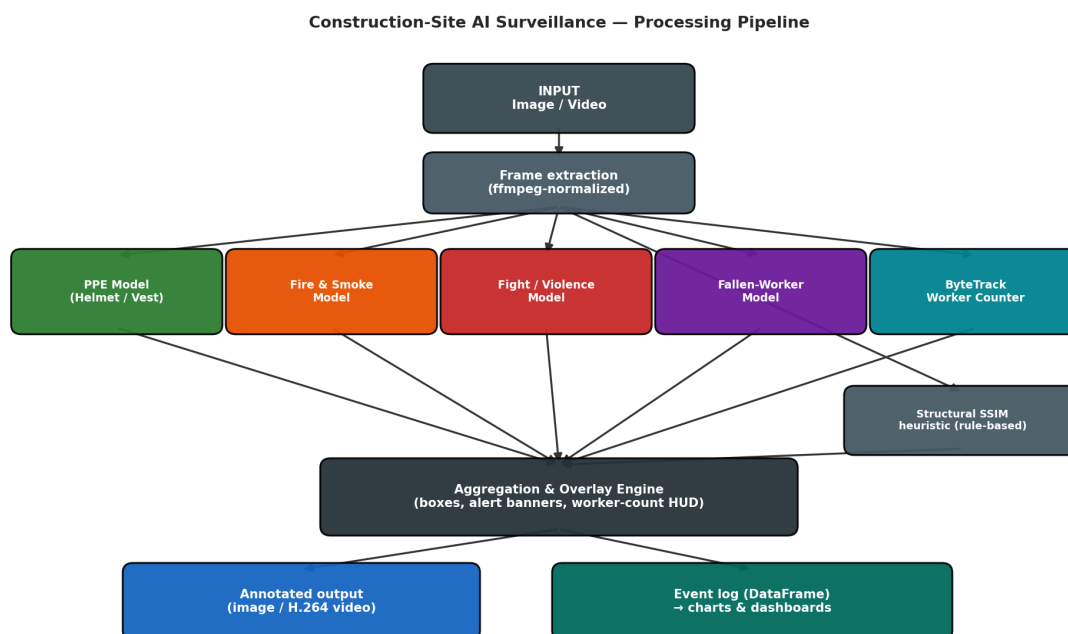


Figure 1 — System processing pipeline, generated directly from the project's own architecture-diagram code.

4. Application Interface

The frontend, branded **AI Surveillance — “See. Detect. Respond.”**, was built as a web application in Visual Studio Code and provides an operational console for site safety staff. Its **Upload & Test** module — the AI Detection Lab shown below — lets a user upload an image or video directly into the browser and see the model's output side-by-side with the original footage, including per-detection confidence scores and severity labels.

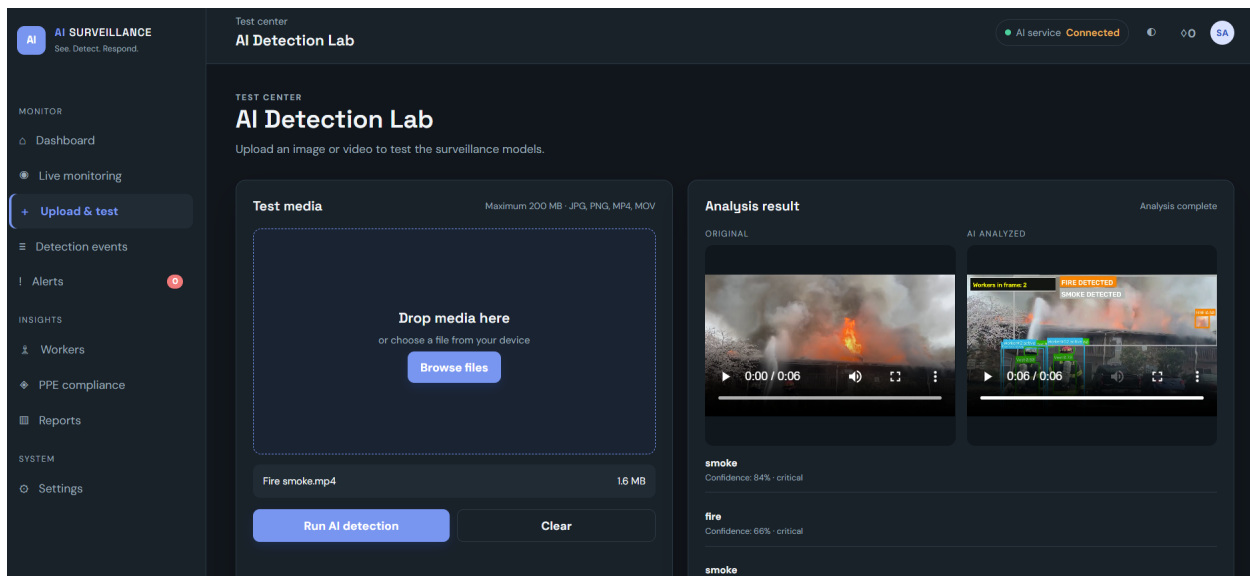


Figure 2 — The AI Detection Lab (Upload & Test) screen of the actual application, showing a fire/smoke test case with live confidence scores.

The left-hand navigation organizes the product into three functional groups: **Monitor** (Dashboard, Live Monitoring, Upload & Test, Detection Events, Alerts), **Insights** (Workers, PPE Compliance, Reports), and **System** (Settings) — separating day-to-day monitoring from historical analysis and configuration.

5. Sample Detection Outputs

The following illustrations show representative output from the PPE and fire/smoke detection modules — each bounding box, label, and confidence score follows the exact visual convention used by the live system: green for compliant, red for a violation or hazard, with per-worker PPE status and a live headcount rendered directly on the frame.

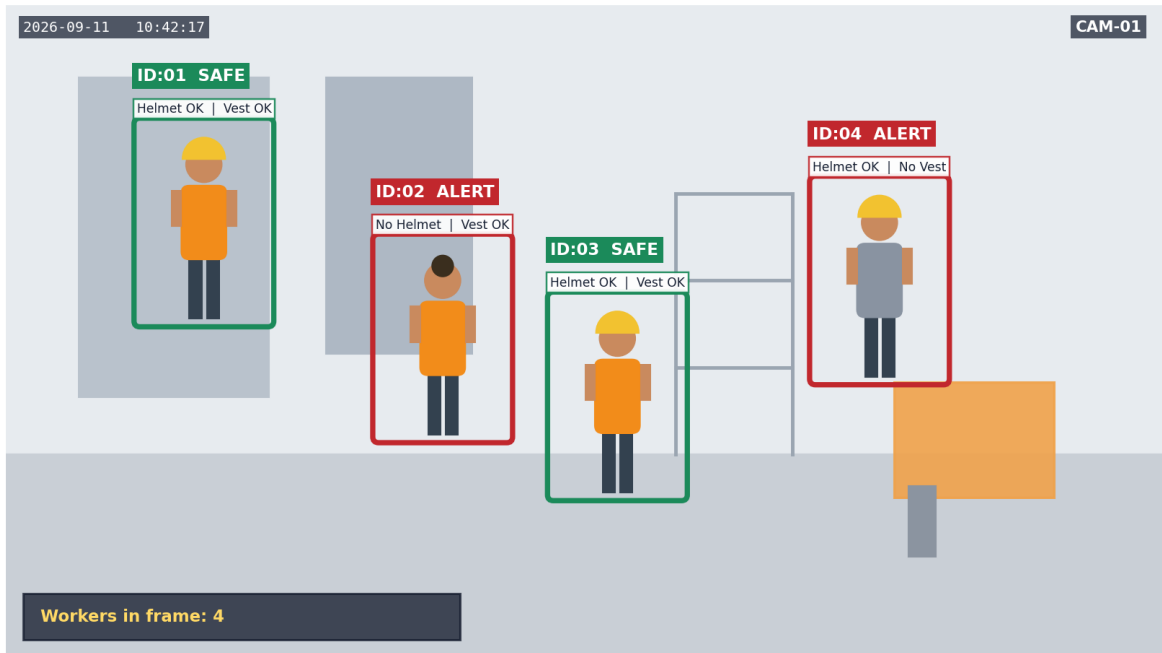


Figure 3 — Illustrative output: worker PPE compliance detection with live headcount overlay.



Figure 4 — Illustrative output: fire and smoke detection with a full-frame critical alert banner.

6. Datasets and Pretrained Models Used

Every detector is powered by a genuine pretrained YOLOv8-family model, sourced and downloaded programmatically from Hugging Face at runtime — no detection module relies on fabricated or simulated weights.

Capability	Model / Source	Dataset Basis & Notes
PPE Compliance (Helmet & Vest)	killuminati1/ construction-ppe-yolov8	YOLOv8, fine-tuned on a labeled construction-site PPE image dataset (19 classes incl. Hard_hat, No-Helmet, Vest, No-Vest, Worker). Reported mAP50 $\approx$ 0.76.
Fire & Smoke	rabahdev/ fire-smoke-yolov8n	YOLOv8n, fine-tuned on the D-Fire dataset (~21,000 labeled fire/smoke images). Reported mAP50 $\approx$ 0.75.
Fight / Altercation	Musawer14/ fight_detection_yolov8	YOLOv8-nano, trained on a labeled fight / no-fight surveillance image dataset. Community-trained; no formally published benchmark.
Fallen / Inactive Worker	melihuzunoglu/ human-fall-detection	YOLOv8/11, trained to distinguish Fallen, Sitting, and Standing postures. Community-trained; no formally published benchmark.
Structural Anomaly	Rule-based heuristic (no external model)	Frame-to-frame Structural Similarity Index (SSIM) against a slowly-adapting reference frame; flags sudden large drops as a possible structural emergency.

Note on transparency: the fight-detection and fallen-worker models are real, working pretrained models, but — unlike the PPE and fire/smoke models — their authors have not published formal accuracy benchmarks. The structural-anomaly detector is deliberately documented as a heuristic rather than a trained classifier, reflecting the current absence of a reliable public pretrained model for building-collapse detection. All three should be validated against site-specific footage before being relied upon for safety-critical decisions.

7. AI Processing Pipeline (Backend)

The core detection engine was developed and benchmarked on Kaggle, taking advantage of free GPU acceleration (NVIDIA T4 / P100) for real-time-capable inference. Development proceeded iteratively: an initial pipeline was built, then hardened against real-world failure modes discovered through hands-on testing with actual uploaded footage — including unreadable video codecs, browser video-preview incompatibility, and out-of-order cell execution in the notebook environment. Key engineering elements include:

- Automatic GPU detection and utilization across all five deep learning models.
- FFmpeg-based video normalization to reliably read a wide range of real-world video codecs and containers.
- H.264 re-encoding of annotated output video for universal browser/media-player compatibility.
- A single, shared detection engine design to keep the analysis notebook and the frontend application logically consistent.
- Per-event-type alert cooldown logic to prevent notification flooding from a single ongoing incident.
- A structured, timestamped event log (CSV) capturing every detection for downstream analytics.
- An analytics suite producing violation breakdowns, event timelines, model benchmark charts, worker-count trends, and a structural-similarity trace — all generated directly from real detection output.

Development & Validation Approach

Rather than assuming generated code was correct, each stage of the pipeline was validated against representative inputs before being finalized — including automated end-to-end dry runs of every processing cell, and visual

inspection of generated diagrams and charts prior to inclusion in the system or this report. This iterative, verify-before-ship approach surfaced and corrected several real defects (e.g. a video-codec compatibility issue and an incorrect diagram-arrow calculation) before they reached end users.

8. Alarm and Email Notification System

The system distinguishes between informational detections (e.g. a single PPE violation) and critical safety events that warrant immediate human attention. The following event types are classified as critical and trigger the full alarm workflow: fire, fight/altercation, fallen or unresponsive worker, and structural anomaly.

- **Visual alert:** a pulsing, high-contrast alarm banner is displayed across the monitoring dashboard.
- **Audible alert:** an on-screen siren indicator activates to draw immediate attention in a live monitoring setting.
- **Email notification:** an email is automatically sent to the designated site owner or safety officer, including an annotated screenshot of the triggering event and a timestamp.
- **Cooldown control:** each alert type is rate-limited so a single ongoing incident (e.g. a 30-second fire) generates one notification rather than one per frame.

9. Frontend Dashboard — Modules and Design

To make the system usable by non-technical site supervisors, the AI Surveillance web application was designed and built in Visual Studio Code, translating the backend's raw detection output into a clear, real-time operational view.

Application Modules

- **Dashboard** — at-a-glance site status and key safety metrics.
- **Live Monitoring** — real-time view across connected camera feeds.
- **Upload & Test (AI Detection Lab)** — upload an image or video and view side-by-side original vs. AI-analyzed output with per-detection confidence scores.
- **Detection Events** — a chronological log of every detection the system has made.
- **Alerts** — critical notifications requiring supervisor attention, with unread count badge.
- **Workers** — a roster view of tracked individuals and their current safety status.
- **PPE Compliance** — aggregated helmet/vest compliance analytics.
- **Reports** — exportable historical safety reporting.
- **Settings** — system configuration and AI service connection status.

The interface follows a clean, professional visual design system with consistent status-color coding (green/amber/red), a live “AI service: Connected” indicator to build user trust in the system's operational state, and a persistent notification bell so no alert goes unnoticed while navigating between modules.

10. Product Requirements Document (PRD)

10.1 Purpose and Objectives

AI-Based Construction Site Surveillance System exists to reduce preventable injuries and fatalities on construction sites by providing continuous, automated safety monitoring that a human supervisor cannot physically sustain alone. The product's primary objectives are:

- Detect PPE non-compliance, fire/smoke, physical altercations, and fallen/inactive workers in real time, from standard camera footage.
- Notify the responsible person immediately when a critical safety event occurs, with enough context (screenshot, timestamp, camera ID) to act without delay.
- Provide a predictive, aggregated view of site risk so supervisors can intervene before an incident occurs, not only after.
- Present all of the above through an interface usable by site supervisors without a technical or data-science background.

10.2 Target Users

User Role	Primary Use of the System
Site Safety Officer	Monitors live alerts, acknowledges incidents, reviews the risk dashboard daily.
Site Supervisor / Manager	Reviews Reports and PPE Compliance analytics; makes staffing and process decisions.
Site Owner / Executive	Receives critical email alerts; reviews high-level safety reporting periodically.
System Administrator	Configures thresholds, camera connections, and notification recipients in Settings.

10.3 Functional Requirements

ID	Requirement
FR-1	The system shall detect helmet and safety-vest presence per worker in each analyzed frame.
FR-2	The system shall detect fire and smoke events and classify them as critical.
FR-3	The system shall detect physical altercations between individuals.
FR-4	The system shall detect a worker who has fallen or has not moved beyond a configurable time threshold.
FR-5	The system shall display a live count of workers visible in the current frame.
FR-6	The system shall flag sudden, large structural changes in the monitored scene as a possible structural emergency.
FR-7	The system shall raise a visible and audible alarm within the interface upon any critical event.

ID	Requirement
FR-8	The system shall send an email notification with an annotated screenshot to a configured recipient upon any critical event, subject to a cooldown period per event type.
FR-9	The system shall compute and display an aggregated site risk score derived from recent detection history.
FR-10	The system shall allow a user to upload an image or video and view analyzed output directly in the browser.
FR-11	The system shall maintain a historical, filterable log of all detections and alerts.

10.4 Non-Functional Requirements

Category	Requirement
Performance	GPU-accelerated inference; the pipeline is designed to process footage at or near real-time frame rates on a single consumer/cloud GPU (e.g. NVIDIA T4).
Reliability	Video input normalization (FFmpeg) to tolerate inconsistent real-world camera formats; explicit error surfacing rather than silent failure at every pipeline stage.
Usability	The interface must be operable by a site supervisor with no machine learning or programming background.
Notification Latency	A critical event shall generate a visible alert and, where configured, an email within seconds of detection.
Transparency	Every detection module's underlying model and confidence score must be visible to the user; heuristic (non-model-based) detectors must be clearly labeled as such.
Extensibility	New camera feeds, sites, or detection categories must be addable without redesigning the core pipeline.

10.5 Success Metrics

- Reduction in mean time-to-notice for critical safety events, compared to manual monitoring.
- PPE compliance rate trend over time at a monitored site (target: measurable improvement after deployment).
- Alert precision — proportion of triggered critical alerts that reflect a genuine, actionable event, as reviewed by site staff.
- Adoption — proportion of flagged incidents acknowledged and acted upon by supervisors via the dashboard.

10.6 Assumptions and Constraints

- Assumes existing site cameras (or uploaded footage) provide sufficiently clear, adequately lit views of monitored areas.
- Assumes a configured email recipient is actively monitored during working hours.
- The structural-anomaly detector is a heuristic and is constrained by the current lack of a labeled, public building-collapse dataset.

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- The fight and fallen-worker models, while functional, are constrained by their community-trained (non-benchmarked) origin and should be tuned per deployment site.

11. Real-World Usefulness and Impact

- **Faster emergency response** — automated detection and instant alerting removes reliance on a human continuously watching every camera feed, reducing the time between an incident occurring and help arriving.
- **Continuous, unbiased PPE compliance monitoring** — every worker in frame is checked consistently, rather than relying on periodic manual spot-checks.
- **Early intervention through risk prediction** — the aggregated site risk score allows supervisors to act on rising risk trends before they culminate in an incident.
- **Auditable safety records** — the structured event log and historical dashboards support regulatory compliance reporting and internal safety audits.
- **Reduced liability and insurance exposure** — demonstrable, continuous safety monitoring can support insurance and regulatory compliance documentation.
- **Scalability** — the same pipeline can be extended to additional cameras, sites, and hazard types without redesigning the core system.
- **Accessible to non-technical staff** — the dashboard is designed for site supervisors and safety officers, not machine learning practitioners, making real-world adoption realistic.

12. Technology Stack

Layer	Technologies Used
AI / Computer Vision	Python, Ultralytics YOLOv8, OpenCV, Hugging Face Hub, PyTorch (GPU-accelerated)
Video Processing	FFmpeg (codec normalization & H.264 encoding)
Data & Analytics	Pandas, Matplotlib, Seaborn, scikit-image (SSIM)
Development / Experimentation	Kaggle Notebooks (GPU: NVIDIA T4 / P100)
Frontend	Web application built in Visual Studio Code
Notifications	SMTP-based automated email alerts with image attachments

13. Limitations and Future Scope

- Structural/collapse detection is currently a rule-based heuristic rather than a trained classifier, reflecting the absence of a reliable public pretrained model for this specific hazard.
- The fight-detection and fallen-worker models are community-trained without formally published accuracy benchmarks and should be validated on site-specific footage before safety-critical deployment.
- Live camera / RTSP ingestion is planned as a direct extension of the existing batch video pipeline.
- Future work includes fine-tuning each model on site-specific footage, adding worker re-identification to improve headcount accuracy under occlusion, and expanding the risk model with historical incident data as it becomes available.

14. Conclusion

AI-Based Construction Site Surveillance System demonstrates a practical, end-to-end application of applied computer vision to a high-stakes, real-world safety problem. By combining multiple genuine pretrained deep learning models, an automated alarm and notification system, predictive risk analytics, and a professional, usable frontend, the project moves beyond a proof-of-concept detector into a coherent system that a construction site could realistically adopt to improve worker safety outcomes.

Appendix A: Glossary of Terms

Term	Definition
PPE	Personal Protective Equipment — safety gear such as a helmet (hard hat) and high-visibility vest.
YOLOv8	“You Only Look Once,” version 8 — a fast, widely-used deep learning architecture for real-time object detection.
mAP50	Mean Average Precision at a 0.50 Intersection-over-Union threshold — a standard accuracy metric for object detection models; higher is better.
SSIM	Structural Similarity Index — a measure of how visually similar two images are, used here to detect sudden structural changes between frames.
Bounding Box	The rectangle a detection model draws around an object it has identified, here shown around workers, fire, or other detected hazards.
Confidence Score	A model's own estimate (0 to 1) of how certain it is that a given detection is correct.
Inference	The process of running a trained model on new data (e.g. a video frame) to produce a prediction.
FFmpeg	An open-source multimedia framework used here to normalize and re-encode video files for reliable processing and playback.
RTSP	Real Time Streaming Protocol — a standard protocol for streaming live video from IP cameras, relevant to planned live-camera support.
Cooldown (alert)	A minimum time interval enforced between repeated notifications for the same alert type, to prevent notification flooding.
Heuristic	A rule-based method (as opposed to a trained machine learning model) used here for structural-anomaly detection.
Risk Score	An aggregated, computed value representing the current overall safety risk level of a monitored site.